
EEE 6778. Applied Machine Learning II



Module 6. Advanced Deep Learning Architectures Review

Multimodal Deep Learning

- Multimodal learning integrates **multiple types of data**, such as images, text, audio, and structured data, to create richer and more contextual AI systems.
- Advanced architectures use **cross-attention mechanisms**, as seen in models like CLIP (Contrastive Language–Image Pre-Training) and Flamingo, enabling one modality to guide the attention mechanism of another.
- **Applications** include medical diagnosis (combining imaging data and clinical notes), autonomous systems (vision, lidar, and GPS), and e-commerce search systems that understand text, images, and user behavior.

Diffusion Models

- Diffusion models are a class of generative models that learn to generate data by **iteratively reversing a process of adding noise**.
- The model starts with pure noise and learns to denoise it step by step, gradually reconstructing the target data distribution.
- This process results in stable training and high-quality sample generation without the mode collapse issues commonly seen in GANs.
 - Models like DALL·E 2, Stable Diffusion, and Google's Imagen have demonstrated the power of diffusion models in generating photorealistic images conditioned on text prompts.

Edge AI and Deep Learning

Edge AI refers to running AI inference directly on devices rather than in the cloud.

- Edge AI is crucial in applications where **immediate responses** are necessary, such as autonomous vehicles, drones, and wearable health devices.
- Deep learning on the edge requires **optimization** techniques like model pruning, quantization (reducing model precision from floating-point to INT8), and the use of lightweight architectures such as MobileNet and EfficientNet.
- **Federated learning** allows models to be trained on edge devices collaboratively without sending sensitive data to centralized servers, maintaining privacy. Examples include smart home assistants like Google Home, which process certain commands locally, and industrial machines that use edge-based object detection for safety monitoring.

Graph Neural Networks

- A graph consists of **nodes** that represent entities and **edges** that represent relationships among them.
- Each node has **features** describing it, and edges can also include features that describe the connection type or strength.
- The main idea is **message passing**, where each node gathers and **aggregates** information from its neighboring nodes.
- Through multiple layers, nodes update their feature representations based on their own features and those of their neighbors.

Recommendation Systems

- Recommendation systems are designed to help users discover relevant content or products.
- Traditional systems are generally based on two major approaches: collaborative filtering and content-based filtering.
 - **Collaborative filtering** analyzes historical interactions between users and items to suggest new items based on similar users' preferences.
 - **Content-based filtering** uses metadata about items, such as categories, descriptions, or other attributes, to recommend items similar to those a user has liked before.



Deep Learning for Recommendation Systems

- Deep learning has revolutionized recommender systems by enabling them to model complex, non-linear relationships between users and items without manual feature engineering.
 - **Neural Collaborative Filtering** replaces simple dot products in matrix factorization with neural networks to learn complex interaction functions.
 - Sequence-based models like **recurrent neural networks (RNNs)** and **Transformers** capture the sequential patterns in user behavior to predict future interests.
 - **Graph Neural Networks (GNNs)** effectively model user-item interaction graphs, enabling more structured recommendations.



And more!

Metalearning in Deep Learning

- Meta-learning, or **learning to learn**, enables models to quickly adapt to new tasks with minimal data.
- **Gradient-based meta-learning** methods like Model-Agnostic Meta-Learning (MAML) aim to find initial model parameters that can be rapidly fine-tuned for new tasks.
- **Metric-based meta-learning methods**, such as Prototypical Networks, focus on learning distance functions that enable few-shot classification.
- **Optimization-based approaches** learn optimizers themselves that adapt quickly to new learning tasks.